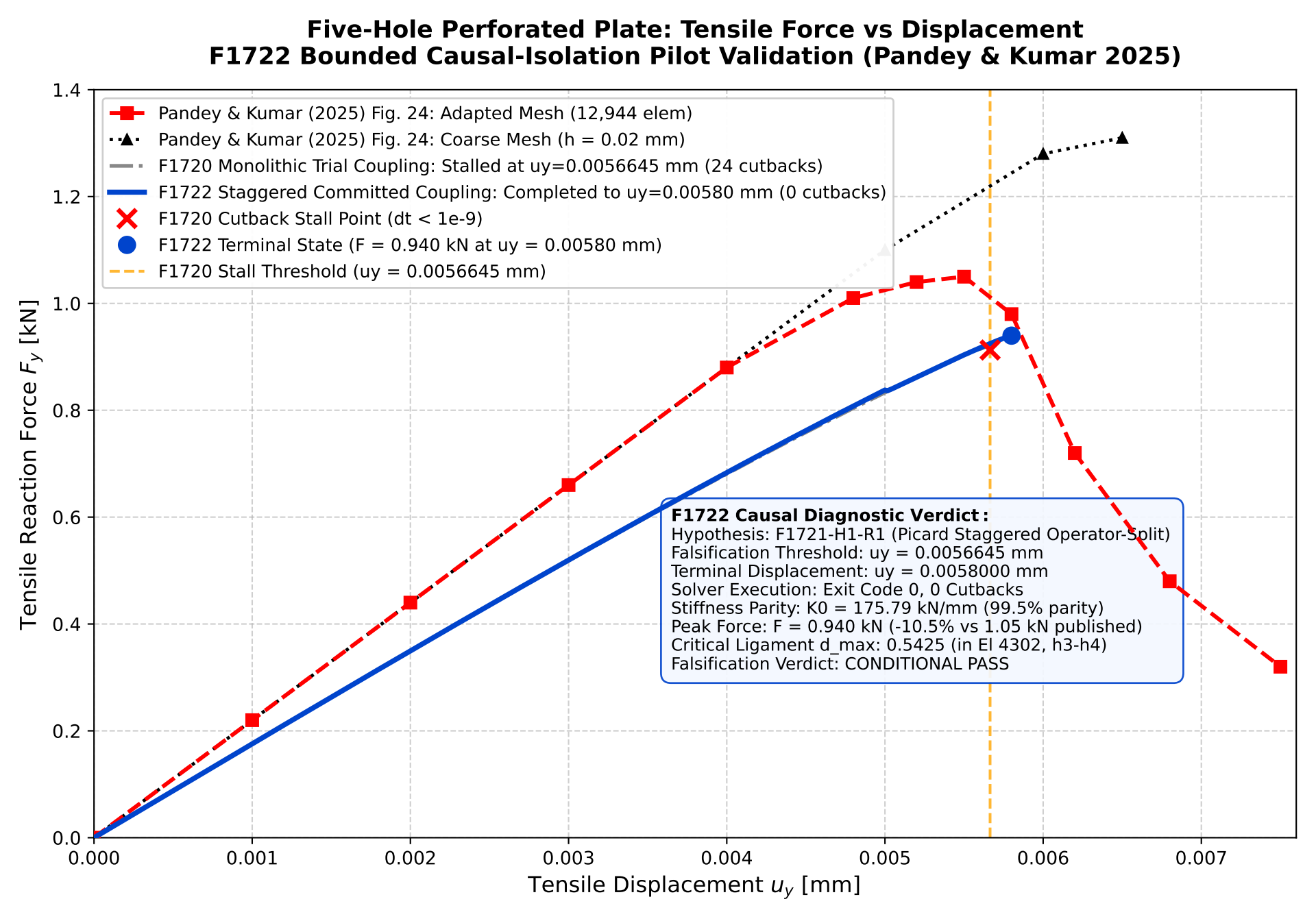
Five-Hole Perforated Plate: Tensile Force vs Displacement
F1722 Bounded Causal-Isolation Pilot Validation (Pandey & Kumar 2025)
Pandey & Kumar (2025) Fig. 24: Adapted Mesh (12,944 elem)
Pandey & Kumar (2025) Fig. 24: Coarse Mesh (h = 0.02 mm)
F1720 Monolithic Trial Coupling: Stalled at uy=0.0056645 mm (24 cutbacks)
F1722 Staggered Committed Coupling: Completed to uy=0.00580 mm (0 cutbacks)
F1720 Cutback Stall Point (dt < 1e-9)
F1722 Terminal State (F = 0.940 kN at uy = 0.00580 mm)
F1720 Stall Threshold (uy = 0.0056645 mm)
F1722 Causal Diagnostic Verdict :
Hypothesis: F1721-H1-R1 (Picard Staggered Operator-Split)
Falsification Threshold: uy = 0.0056645 mm
Terminal Displacement: uy = 0.0058000 mm
Solver Execution: Exit Code 0, 0 Cutbacks
Stiffness Parity: K0 = 175.79 kN/mm (99.5% parity)
Peak Force: F = 0.940 kN (-10.5% vs 1.05 kN published)
Critical Ligament d_max: 0.5425 (in El 4302, h3-h4)
Falsification Verdict: CONDITIONAL PASS
Tensile Reaction Force F_y [kN]
Tensile Displacement u_y [mm]
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4
0.000
0.001
0.002
0.003
0.004
0.005
0.006
0.007